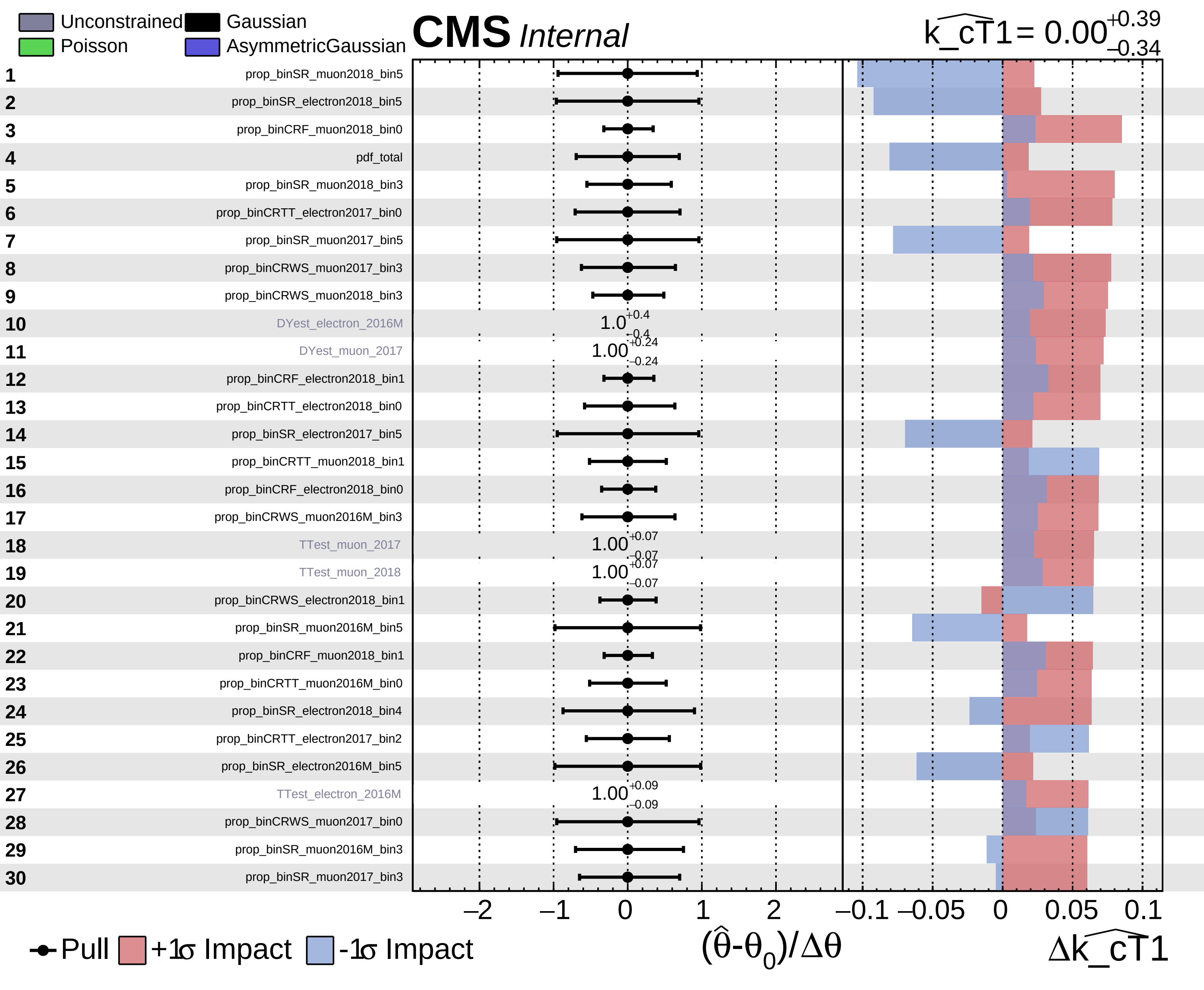
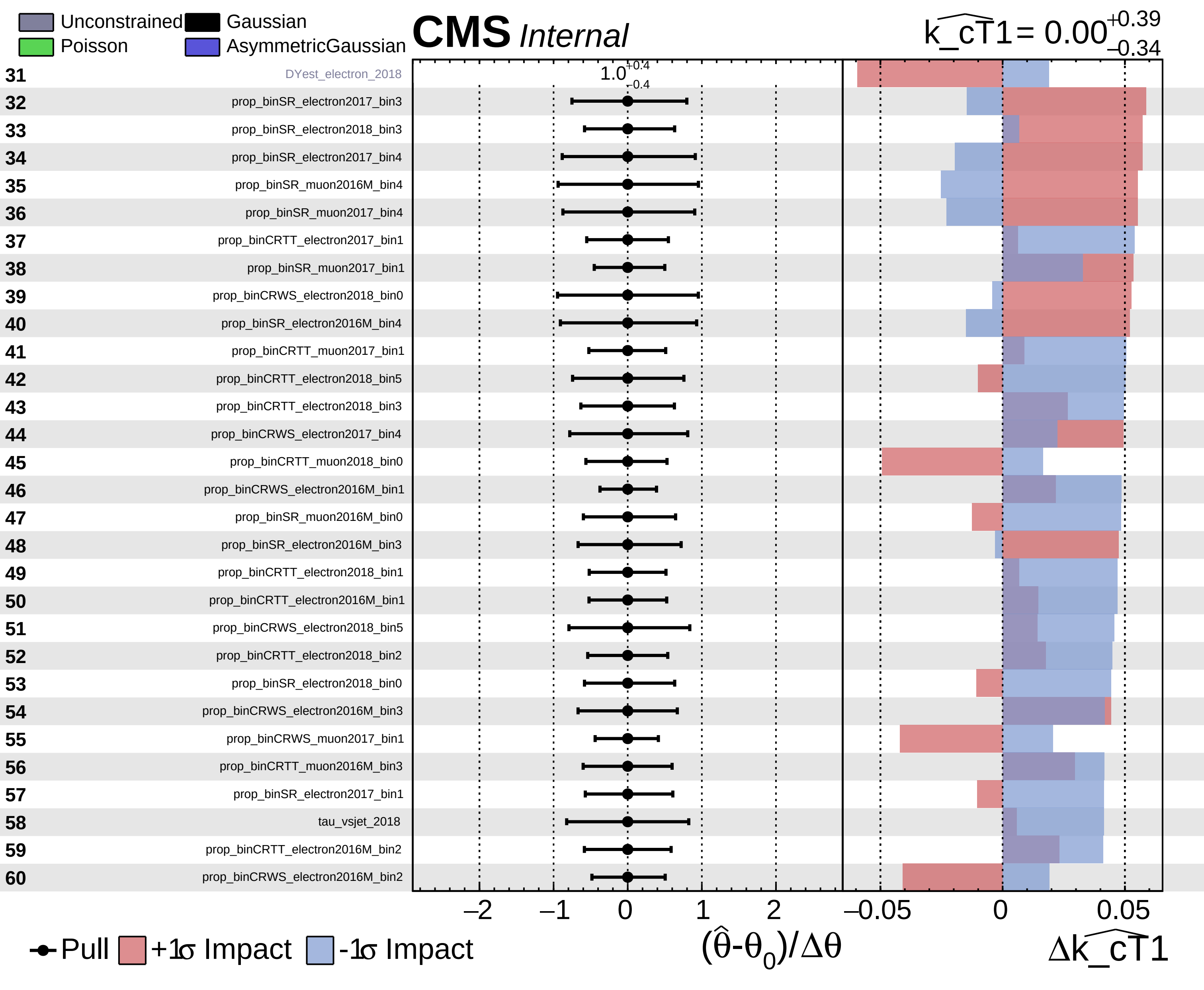
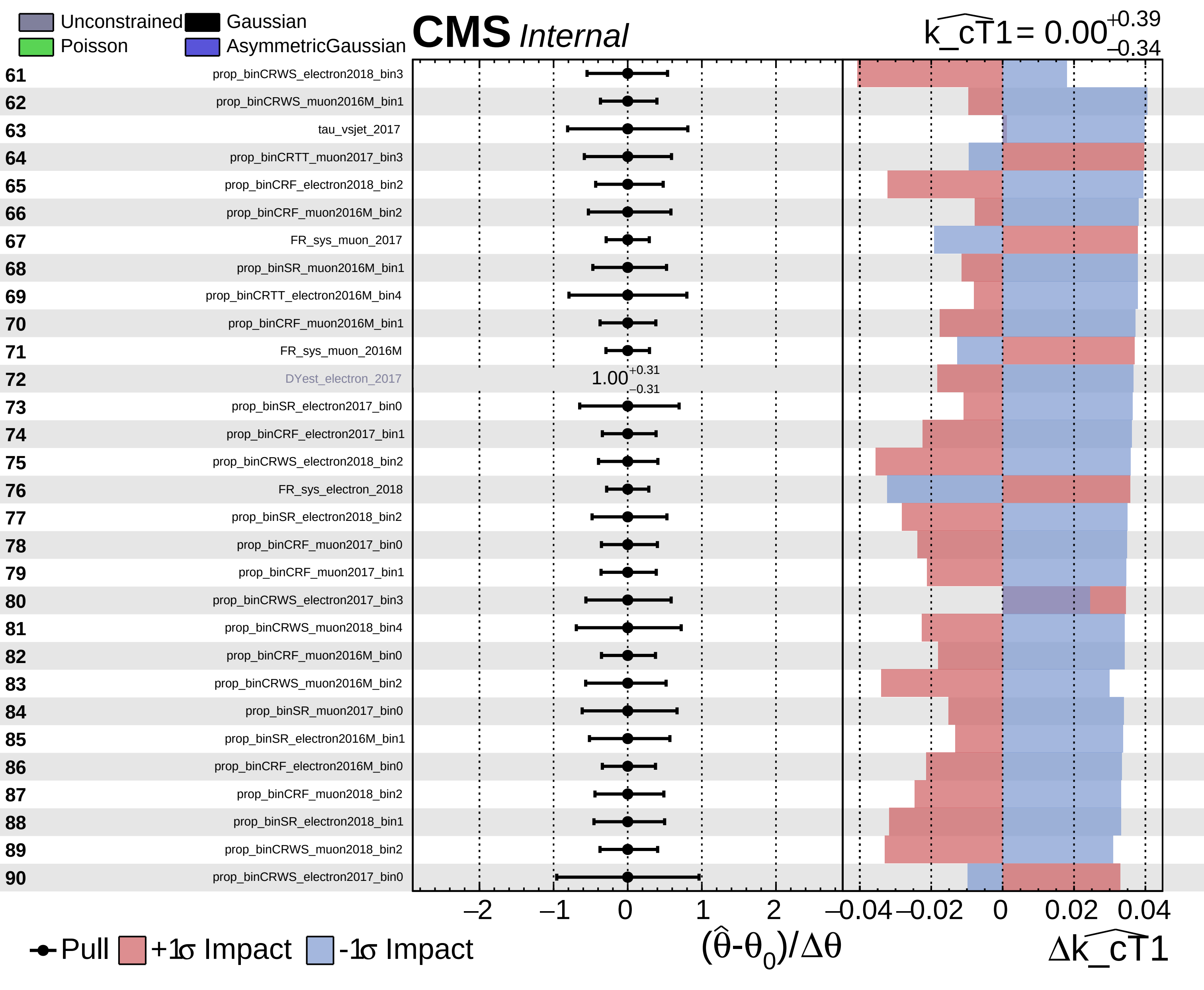


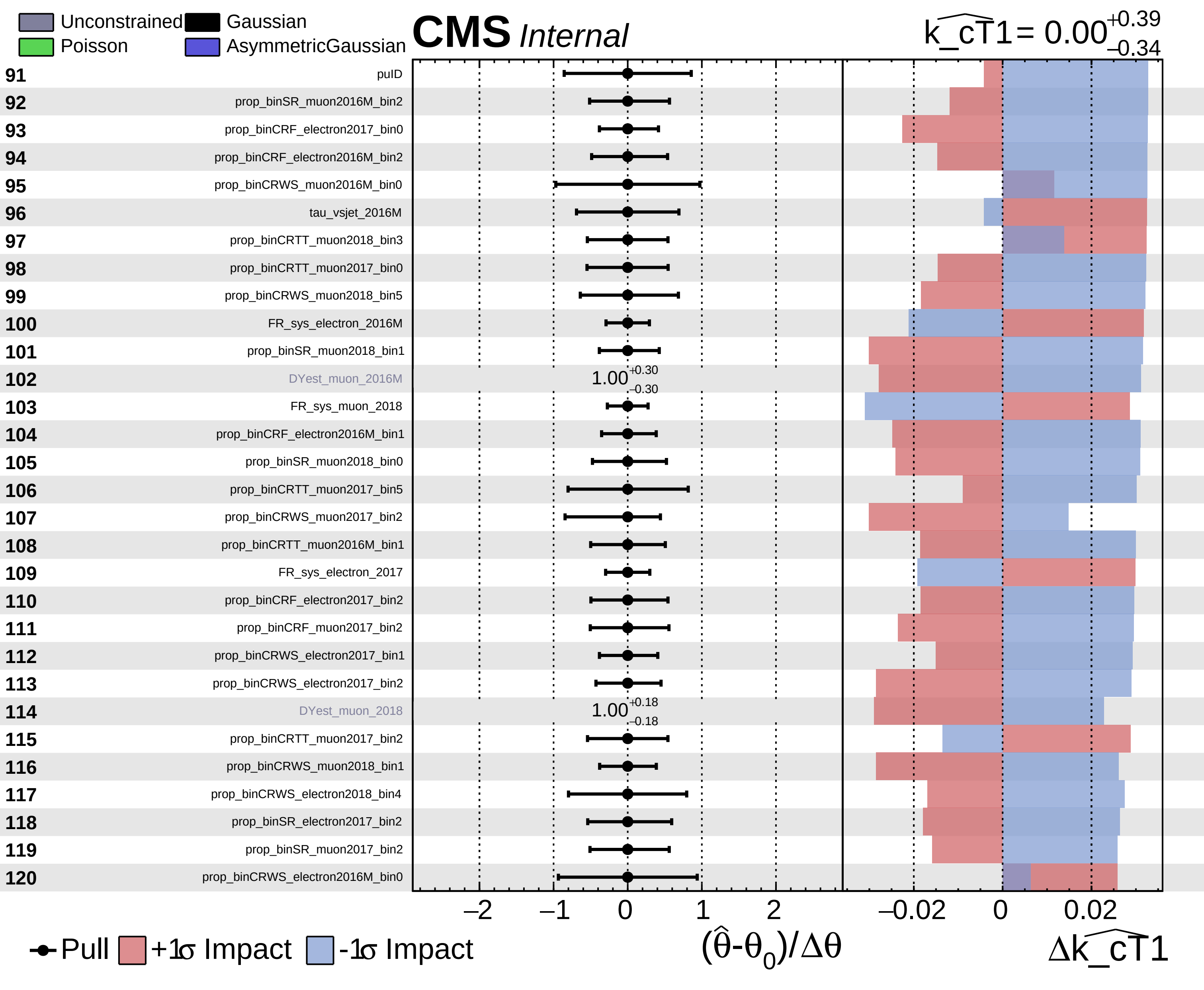
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = 0.00^{+0.39}_{-0.34}$
1 prop_binSR_muon2018_bin5
2 prop_binSR_electron2018_bin5
3 prop_binCRF_muon2018_bin0
4 pdf_total
5 prop_binSR_muon2018_bin3
6 prop_binCRTT_electron2017_bin0
7 prop_binSR_muon2017_bin5
8 prop_binCRWS_muon2017_bin3
9 prop_binCRWS_muon2018_bin3
10 DYest_electron_2016M
$1.0^{+0.4}_{-0.4}$
11 DYest_muon_2017
$1.00^{+0.24}_{-0.24}$
12 prop_binCRF_electron2018_bin1
13 prop_binCRTT_electron2018_bin0
14 prop_binSR_electron2017_bin5
15 prop_binCRTT_muon2018_bin1
16 prop_binCRF_electron2018_bin0
17 prop_binCRWS_muon2016M_bin3
18 TTest_muon_2017
$1.00^{+0.07}_{-0.07}$
19 TTest_muon_2018
$1.00^{+0.07}_{-0.07}$
20 prop_binCRWS_electron2018_bin1
21 prop_binSR_muon2016M_bin5
22 prop_binCRF_muon2018_bin1
23 prop_binCRTT_muon2016M_bin0
24 prop_binSR_electron2018_bin4
25 prop_binCRTT_electron2017_bin2
26 prop_binSR_electron2016M_bin5
27 TTest_electron_2016M
$1.00^{+0.09}_{-0.09}$
28 prop_binCRWS_muon2017_bin0
29 prop_binSR_muon2016M_bin3
30 prop_binSR_muon2017_bin3
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$



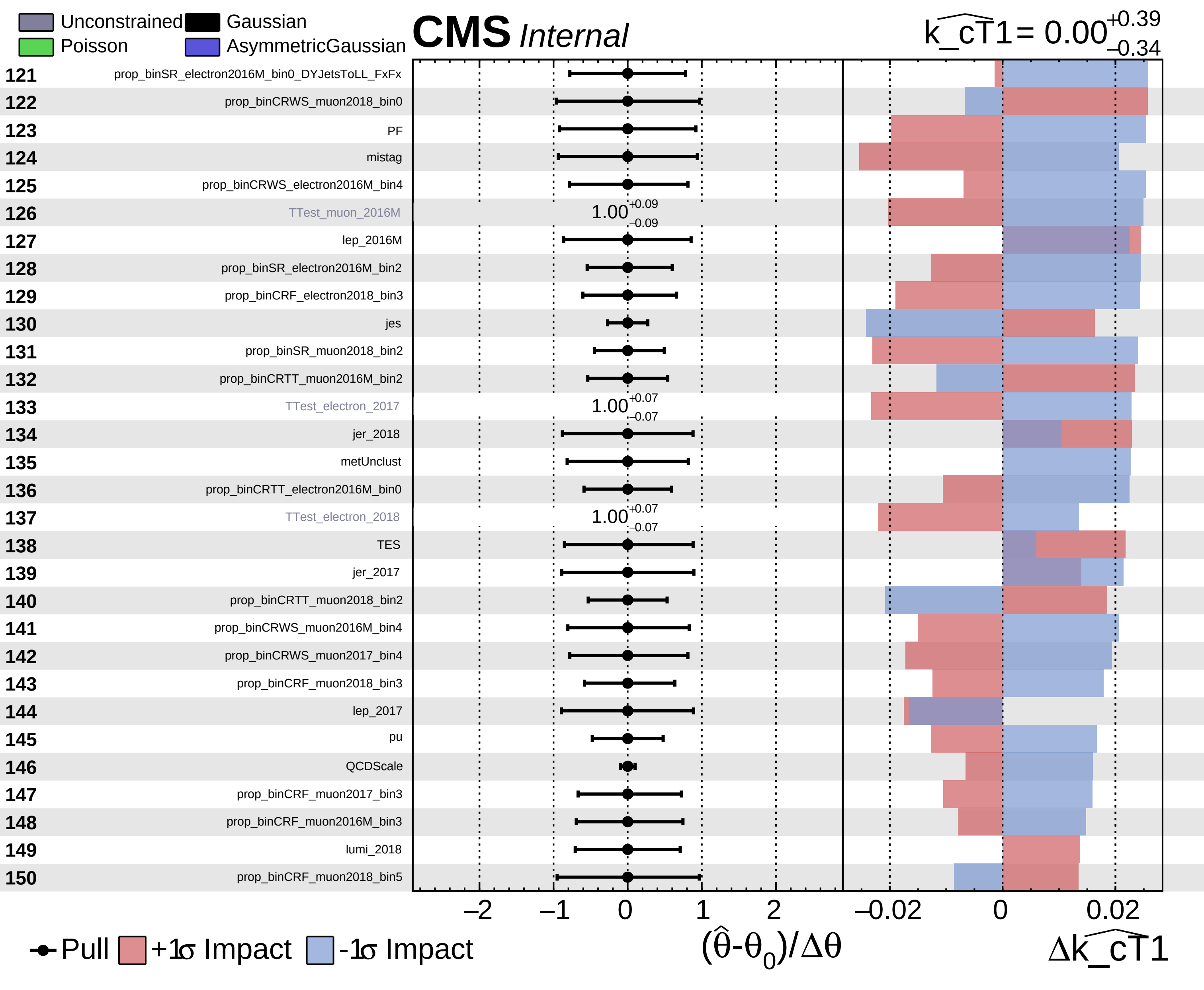
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.39 −0.34
31 DYest_electron_2018
1.0 +0.4 −0.4
32 prop_binSR_electron2017_bin3
33 prop_binSR_electron2018_bin3
34 prop_binSR_electron2017_bin4
35 prop_binSR_muon2016M_bin4
36 prop_binSR_muon2017_bin4
37 prop_binCRTT_electron2017_bin1
38 prop_binSR_muon2017_bin1
39 prop_binCRWS_electron2018_bin0
40 prop_binSR_electron2016M_bin4
41 prop_binCRTT_muon2017_bin1
42 prop_binCRTT_electron2018_bin5
43 prop_binCRTT_electron2018_bin3
44 prop_binCRWS_electron2017_bin4
45 prop_binCRTT_muon2018_bin0
46 prop_binCRWS_electron2016M_bin1
47 prop_binSR_muon2016M_bin0
48 prop_binSR_electron2016M_bin3
49 prop_binCRTT_electron2018_bin1
50 prop_binCRTT_electron2016M_bin1
51 prop_binCRWS_electron2018_bin5
52 prop_binCRTT_electron2018_bin2
53 prop_binSR_electron2018_bin0
54 prop_binCRWS_electron2016M_bin3
55 prop_binCRWS_muon2017_bin1
56 prop_binCRTT_muon2016M_bin3
57 prop_binSR_electron2017_bin1
58 tau_vsjet_2018
59 prop_binCRTT_electron2016M_bin2
60 prop_binCRWS_electron2016M_bin2
−2
−1
0
1
2
−0.05
0
0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT1

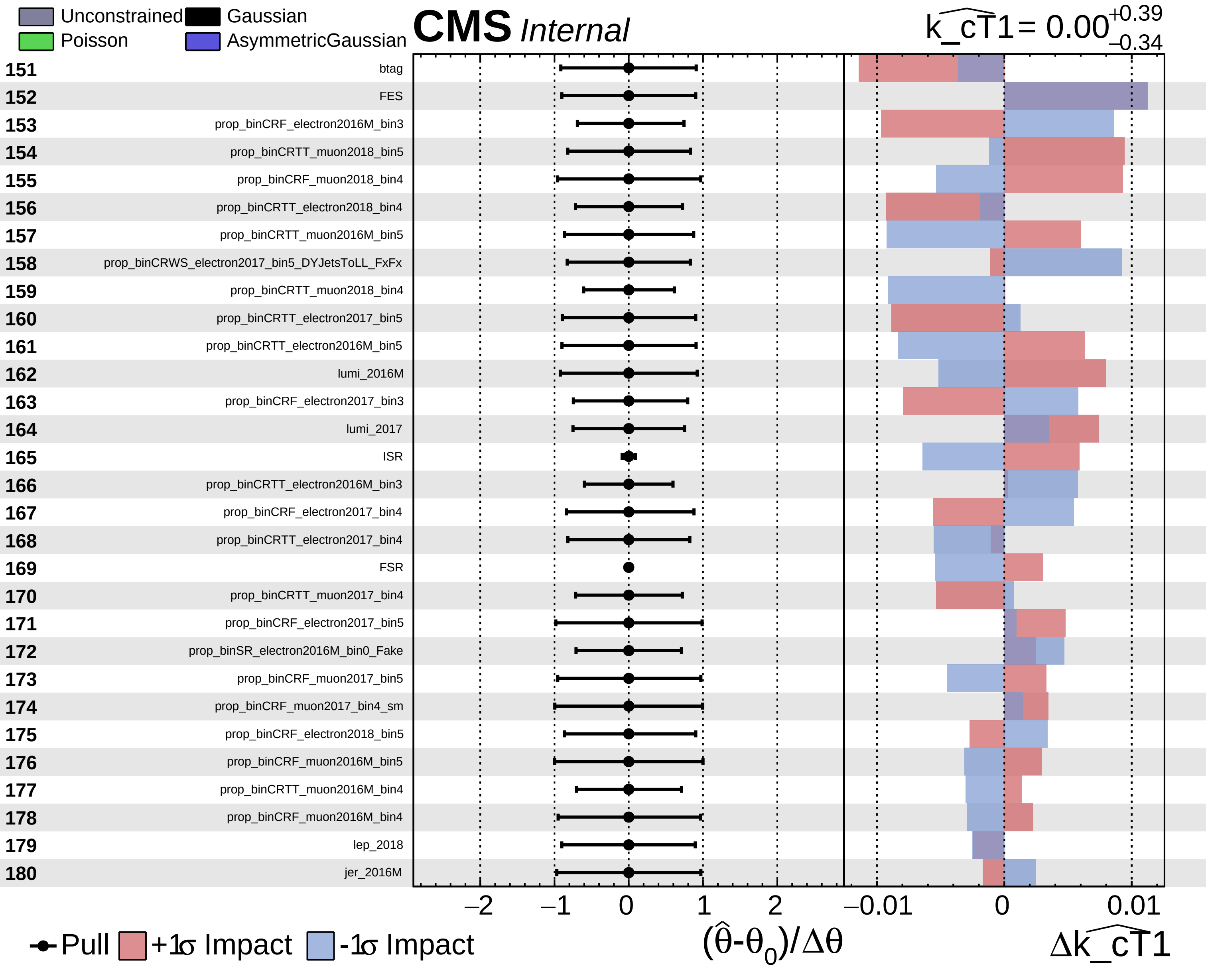


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.39 −0.34
61 prop_binCRWS_electron2018_bin3
62 prop_binCRWS_muon2016M_bin1
63 tau_vsjet_2017
64 prop_binCRTT_muon2017_bin3
65 prop_binCRF_electron2018_bin2
66 prop_binCRF_muon2016M_bin2
67 FR_sys_muon_2017
68 prop_binSR_muon2016M_bin1
69 prop_binCRTT_electron2016M_bin4
70 prop_binCRF_muon2016M_bin1
71 FR_sys_muon_2016M
72 DYest_electron_2017
1.00 +0.31 −0.31
73 prop_binSR_electron2017_bin0
74 prop_binCRF_electron2017_bin1
75 prop_binCRWS_electron2018_bin2
76 FR_sys_electron_2018
77 prop_binSR_electron2018_bin2
78 prop_binCRF_muon2017_bin0
79 prop_binCRF_muon2017_bin1
80 prop_binCRWS_electron2017_bin3
81 prop_binCRWS_muon2018_bin4
82 prop_binCRF_muon2016M_bin0
83 prop_binCRWS_muon2016M_bin2
84 prop_binSR_muon2017_bin0
85 prop_binSR_electron2016M_bin1
86 prop_binCRF_electron2016M_bin0
87 prop_binCRF_muon2018_bin2
88 prop_binSR_electron2018_bin1
89 prop_binCRWS_muon2018_bin2
90 prop_binCRWS_electron2017_bin0
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT1

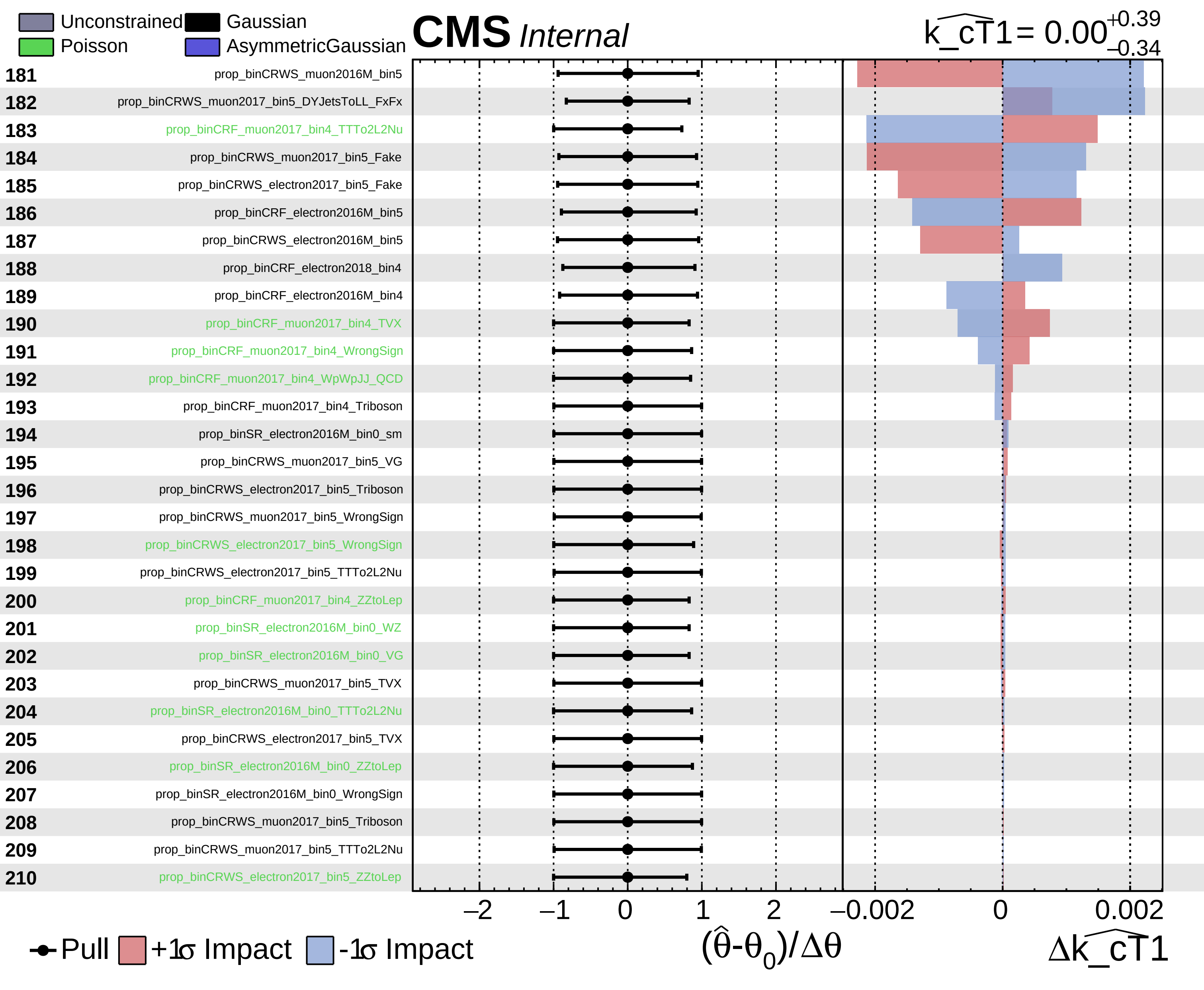


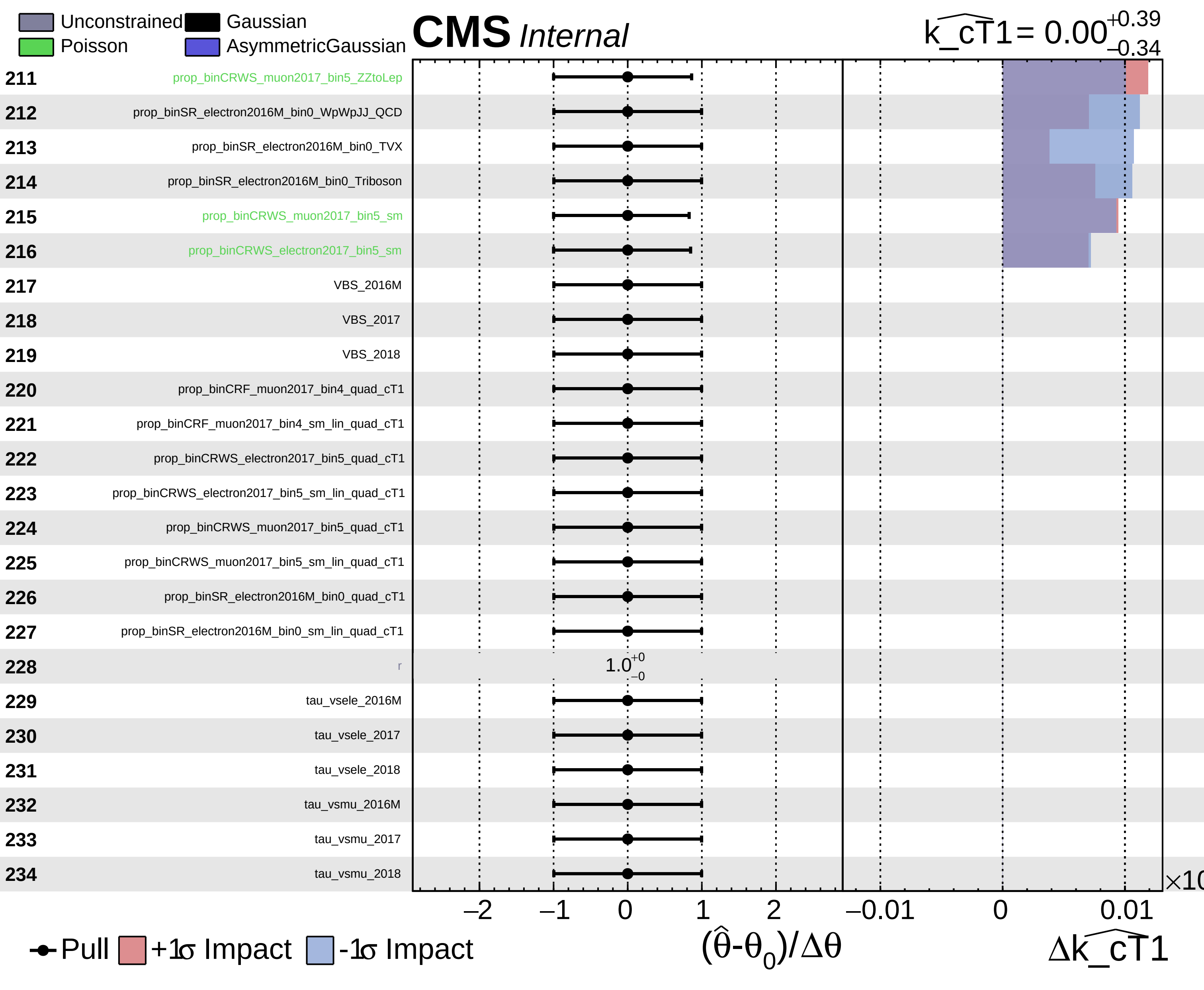
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.39 −0.34
91 puID
92 prop_binSR_muon2016M_bin2
93 prop_binCRF_electron2017_bin0
94 prop_binCRF_electron2016M_bin2
95 prop_binCRWS_muon2016M_bin0
96 tau_vsjet_2016M
97 prop_binCRTT_muon2018_bin3
98 prop_binCRTT_muon2017_bin0
99 prop_binCRWS_muon2018_bin5
100 FR_sys_electron_2016M
101 prop_binSR_muon2018_bin1
102 DYest_muon_2016M
1.00 +0.30 −0.30
103 FR_sys_muon_2018
104 prop_binCRF_electron2016M_bin1
105 prop_binSR_muon2018_bin0
106 prop_binCRTT_muon2017_bin5
107 prop_binCRWS_muon2017_bin2
108 prop_binCRTT_muon2016M_bin1
109 FR_sys_electron_2017
110 prop_binCRF_electron2017_bin2
111 prop_binCRF_muon2017_bin2
112 prop_binCRWS_electron2017_bin1
113 prop_binCRWS_electron2017_bin2
114 DYest_muon_2018
1.00 +0.18 −0.18
115 prop_binCRTT_muon2017_bin2
116 prop_binCRWS_muon2018_bin1
117 prop_binCRWS_electron2018_bin4
118 prop_binSR_electron2017_bin2
119 prop_binSR_muon2017_bin2
120 prop_binCRWS_electron2016M_bin0
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT1


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.39 −0.34
121 prop_binSR_electron2016M_bin0_DYJetsToLL_FxFx
122 prop_binCRWS_muon2018_bin0
123 PF
124 mistag
125 prop_binCRWS_electron2016M_bin4
126 TTest_muon_2016M 1.00 +0.09 −0.09
127 lep_2016M
128 prop_binSR_electron2016M_bin2
129 prop_binCRF_electron2018_bin3
130 jes
131 prop_binSR_muon2018_bin2
132 prop_binCRTT_muon2016M_bin2
133 TTest_electron_2017 1.00 +0.07 −0.07
134 jer_2018
135 metUnclust
136 prop_binCRTT_electron2016M_bin0
137 TTest_electron_2018 1.00 +0.07 −0.07
138 TES
139 jer_2017
140 prop_binCRTT_muon2018_bin2
141 prop_binCRWS_muon2016M_bin4
142 prop_binCRWS_muon2017_bin4
143 prop_binCRF_muon2018_bin3
144 lep_2017
145 pu
146 QCDScale
147 prop_binCRF_muon2017_bin3
148 prop_binCRF_muon2016M_bin3
149 lumi_2018
150 prop_binCRF_muon2018_bin5
−2 −1 0 1 2
−0.02 0 0.02
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.39 −0.34
151 btag
152 FES
153 prop_binCRF_electron2016M_bin3
154 prop_binCRTT_muon2018_bin5
155 prop_binCRF_muon2018_bin4
156 prop_binCRTT_electron2018_bin4
157 prop_binCRTT_muon2016M_bin5
158 prop_binCRWS_electron2017_bin5_DYJetsToLL_FxFx
159 prop_binCRTT_muon2018_bin4
160 prop_binCRTT_electron2017_bin5
161 prop_binCRTT_electron2016M_bin5
162 lumi_2016M
163 prop_binCRF_electron2017_bin3
164 lumi_2017
165 ISR
166 prop_binCRTT_electron2016M_bin3
167 prop_binCRF_electron2017_bin4
168 prop_binCRTT_electron2017_bin4
169 FSR
170 prop_binCRTT_muon2017_bin4
171 prop_binCRF_electron2017_bin5
172 prop_binSR_electron2016M_bin0_Fake
173 prop_binCRF_muon2017_bin5
174 prop_binCRF_muon2017_bin4_sm
175 prop_binCRF_electron2018_bin5
176 prop_binCRF_muon2016M_bin5
177 prop_binCRTT_muon2016M_bin4
178 prop_binCRF_muon2016M_bin4
179 lep_2018
180 jer_2016M
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.39 −0.34
181 prop_binCRWS_muon2016M_bin5
182 prop_binCRWS_muon2017_bin5_DYJetsToLL_FxFx
183 prop_binCRF_muon2017_bin4_TTTo2L2Nu
184 prop_binCRWS_muon2017_bin5_Fake
185 prop_binCRWS_electron2017_bin5_Fake
186 prop_binCRF_electron2016M_bin5
187 prop_binCRWS_electron2016M_bin5
188 prop_binCRF_electron2018_bin4
189 prop_binCRF_electron2016M_bin4
190 prop_binCRF_muon2017_bin4_TVX
191 prop_binCRF_muon2017_bin4_WrongSign
192 prop_binCRF_muon2017_bin4_WpWpJJ_QCD
193 prop_binCRF_muon2017_bin4_Triboson
194 prop_binSR_electron2016M_bin0_sm
195 prop_binCRWS_muon2017_bin5_VG
196 prop_binCRWS_electron2017_bin5_Triboson
197 prop_binCRWS_muon2017_bin5_WrongSign
198 prop_binCRWS_electron2017_bin5_WrongSign
199 prop_binCRWS_electron2017_bin5_TTTo2L2Nu
200 prop_binCRF_muon2017_bin4_ZZtoLep
201 prop_binSR_electron2016M_bin0_WZ
202 prop_binSR_electron2016M_bin0_VG
203 prop_binCRWS_muon2017_bin5_TVX
204 prop_binSR_electron2016M_bin0_TTTo2L2Nu
205 prop_binCRWS_electron2017_bin5_TVX
206 prop_binSR_electron2016M_bin0_ZZtoLep
207 prop_binSR_electron2016M_bin0_WrongSign
208 prop_binCRWS_muon2017_bin5_Triboson
209 prop_binCRWS_muon2017_bin5_TTTo2L2Nu
210 prop_binCRWS_electron2017_bin5_ZZtoLep
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.39 −0.34
211 prop_binCRWS_muon2017_bin5_ZZtoLep
212 prop_binSR_electron2016M_bin0_WpWpJJ_QCD
213 prop_binSR_electron2016M_bin0_TVX
214 prop_binSR_electron2016M_bin0_Triboson
215 prop_binCRWS_muon2017_bin5_sm
216 prop_binCRWS_electron2017_bin5_sm
217 VBS_2016M
218 VBS_2017
219 VBS_2018
220 prop_binCRF_muon2017_bin4_quad_cT1
221 prop_binCRF_muon2017_bin4_sm_lin_quad_cT1
222 prop_binCRWS_electron2017_bin5_quad_cT1
223 prop_binCRWS_electron2017_bin5_sm_lin_quad_cT1
224 prop_binCRWS_muon2017_bin5_quad_cT1
225 prop_binCRWS_muon2017_bin5_sm_lin_quad_cT1
226 prop_binSR_electron2016M_bin0_quad_cT1
227 prop_binSR_electron2016M_bin0_sm_lin_quad_cT1
228 r
1.0 +0 −0
229 tau_vsele_2016M
230 tau_vsele_2017
231 tau_vsele_2018
232 tau_vsmu_2016M
233 tau_vsmu_2017
234 tau_vsmu_2018
−2 −1 0 1 2
−0.01 0 0.01
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1